

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

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QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

The reward values increase gradually from **10 to 45** as the number of training episodes increases. Reward represents how well the RL agent performs its task. A higher reward generally means that the agent is selecting more appropriate actions and achieving better outcomes.

In the beginning, the agent has limited knowledge about the environment, so it takes many exploratory or incorrect actions. Therefore, the reward is low. As training continues, the agent learns from previous experiences and improves its **policy**, which determines the actions taken in different states.

The gradual increase from **10 to 45** shows that the agent is learning successfully. If the reward eventually becomes stable around a high value, it indicates that the agent has reached a relatively stable policy.

Interpretation:

- Reward = 10 → poor initial performance.
- Reward gradually increases → continuous learning.
- Reward = 45 → significantly improved performance.
- Stable high reward → possible policy convergence.

Conclusion: The increasing reward demonstrates **improved learning performance, better decision-making, and policy improvement**. If the reward remains stable at a high level, the agent may have learned a near-optimal policy

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

The epsilon-greedy strategy balances **exploration and exploitation**. The value of ϵ determines the probability that the agent chooses a random action.

For $\epsilon = 0.1$, the agent explores only about 10% of the time and exploits the best-known action about 90% of the time. Therefore, it can make use of its learned knowledge and obtain a high cumulative reward.

For $\epsilon = 0.3$, exploration increases to approximately 30%. This allows the agent to discover new actions but also causes it to occasionally select actions that are not currently optimal.

For $\epsilon = 0.5$, the agent explores approximately half of the time. Consequently, it performs many random actions and spends less time exploiting its learned policy. This results in the lowest cumulative reward in the given experiment.

Epsilon	Exploration	Exploitation	Performance
0.1	Low	High	Highest reward
0.3	Moderate	Moderate	Medium reward
0.5	High	Low	Lowest reward

Conclusion: Since $\epsilon = 0.1$ produces the highest cumulative reward, it provides the best performance in this scenario. It gives the agent enough exploration while emphasizing exploitation of the learned policy.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

Q-Learning and SARSA are temporal-difference reinforcement learning algorithms, but they learn policies differently.

Q-Learning is off-policy, meaning it learns the value of the best possible next action regardless of the action actually selected by the current policy.

SARSA is on-policy, meaning it learns using the action that the current policy actually chooses.

Suppose the average rewards are observed as:

Episodes	Q-Learning	SARSA
100	Higher	Lower
200	Higher	Lower
300	Higher	Lower

The consistent higher reward of Q-Learning indicates that it is finding more rewarding action sequences in this environment.

If Q-Learning reaches high and stable rewards earlier than SARSA, it suggests faster convergence. SARSA may learn more cautiously because it considers the actual exploratory actions taken by the policy.

Conclusion: Based on the observed results, **Q-Learning performs better and appears to converge faster than SARSA**, because it consistently obtains higher rewards throughout training.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

The discount factor, represented by γ (**gamma**), determines how much importance the agent gives to future rewards.

A low discount factor means the agent focuses mainly on immediate rewards. A high discount factor means the agent considers future rewards more strongly.

- $\gamma = 0.2$: The agent strongly prefers immediate rewards.
- $\gamma = 0.5$: The agent considers both immediate and future rewards moderately.

- $\gamma = 0.9$: The agent gives substantial importance to long-term rewards.

Since the average reward increases as γ increases, the environment appears to benefit from long-term planning.

For example, an action may provide a small reward immediately but lead to a much larger reward later. With $\gamma = 0.9$, the agent is more likely to choose such an action.

Conclusion: $\gamma = 0.9$ is the most suitable among the given values because it considers long-term consequences and produces the highest average reward. Therefore, it provides better learning performance in this scenario.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data.

(5 Marks)

The success rate increases from **45% at 50 episodes to 92% at 200 episodes**. This is a substantial improvement of **47 percentage points**.

At 50 episodes, the agent successfully completes the task only 45% of the time. This indicates that the agent is still learning and frequently makes incorrect decisions.

As the number of episodes increases, the agent receives more experience and improves its understanding of the environment. By 200 episodes, the success rate reaches 92%, showing that the agent can successfully complete the task in most situations.

The trend can be interpreted as:

45% → 60% → 75% → 85% → 92%

This upward trend indicates that the learned policy is becoming more effective.

However, a 92% success rate does not necessarily prove mathematical optimality. To claim an optimal policy, the result should normally be compared with the theoretical maximum or evaluated over additional episodes.

Conclusion: The increase from **45% to 92%** demonstrates strong policy improvement and indicates that the agent has learned a **highly effective, near-optimal policy**.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability.

(5 Marks)

The learning rate controls how much the agent changes its learned values after receiving new information.

A very small learning rate makes updates slow, while a very large learning rate can make the learning process unstable.

For **learning rate = 0.01**, the agent makes small updates. This generally produces stable learning but can require many episodes to converge.

For **learning rate = 0.05**, the updates are moderate. The agent can learn at a reasonable speed while maintaining good stability.

For **learning rate = 0.1**, the updates are large. This can increase the speed of learning initially, but the agent may overshoot useful values and experience fluctuations.

Learning Rate	Learning Speed	Stability
0.01	Slow	High
0.05	Moderate/Fast	High
0.1	Fast	Lower

Since the agent achieves the highest stability at **0.05**, this value provides a good balance between rapid learning and stable convergence.

Conclusion: Learning rate = 0.05 is the best choice because it provides a good trade-off between **learning speed, stability, and convergence**.

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

(5 Marks)

The three exploration strategies select actions differently.

ϵ -greedy: The agent usually chooses the best-known action but occasionally chooses a random action. This provides a simple balance between exploration and exploitation.

Softmax: Actions are selected probabilistically according to their estimated values. Better actions have higher probabilities, while weaker actions still have some probability of being selected.

Random exploration: Actions are selected randomly without strongly considering the learned action values. Therefore, the agent may repeatedly choose poor actions.

If ϵ -greedy produces the highest cumulative reward, it indicates that the strategy is effectively exploiting learned information while maintaining sufficient exploration.

Strategy	Main Advantage	Limitation
ϵ -greedy	Simple exploration/exploitation balance	Random exploration
Softmax	Value-based exploration	Sensitive to temperature
Random	High exploration	Poor exploitation

In this experiment, ϵ -greedy achieves the highest reward because it spends most of its time selecting the best-known actions.

Conclusion: ϵ -greedy performs best in this scenario because it provides an effective balance between exploration and exploitation and makes better use of learned knowledge.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

Initially, the RL agent takes approximately **50 steps** to reach the goal. After training for **300 episodes**, the number of steps decreases to **15**.

A lower number of steps indicates that the agent has learned a more efficient route. Initially, the agent may move randomly, take unnecessary paths, or encounter dead ends. Through repeated interaction, it learns which actions lead toward the goal.

The reduction is:

50 steps $\rightarrow$ 15 steps

This represents a reduction of **35 steps**, or approximately **70% fewer steps**.

This is strong evidence that the agent has improved its decision-making and learned a much shorter path.

However, convergence to an optimal policy should be confirmed by observing whether the number of steps remains around 15 for additional episodes.

Conclusion: The reduction from **50 to 15 steps** indicates significant policy improvement and suggests convergence toward an efficient or near-optimal policy. Stable performance over additional episodes would provide stronger evidence of convergence.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

(5 Marks)

The discount factor controls the importance of future rewards in SARSA.

With $\gamma = 0.4$, the agent gives relatively high importance to immediate rewards and less importance to future consequences.

With $\gamma = 0.7$, the agent considers future rewards more strongly while still giving considerable importance to immediate rewards.

With $\gamma = 0.95$, the agent strongly values future rewards. It is therefore more likely to select actions that may provide smaller immediate rewards but result in greater long-term benefits.

Since the highest long-term reward is achieved with $\gamma = 0.95$, the environment favors long-term planning.

For example, an agent may have two choices:

- Action A $\rightarrow$ immediate reward = 10, future reward = small.
- Action B $\rightarrow$ immediate reward = 2, future reward = large.

A high γ makes Action B more attractive if its long-term benefit is greater.

Conclusion: $\gamma = 0.95$ is the most effective value because it strongly considers future rewards and helps the SARSA agent learn a better long-term strategy.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

(5 Marks)

In the early training episodes, the agent's rewards fluctuate significantly. This is expected because the agent has limited knowledge of the environment and is actively exploring different actions.

For example, the reward trend may look like:

20 $\rightarrow$ 35 $\rightarrow$ 15 $\rightarrow$ 40 $\rightarrow$ 25 $\rightarrow$ 45 $\rightarrow$ 42 $\rightarrow$ 44 $\rightarrow$ 43

The fluctuations indicate that the agent is experimenting and learning which actions produce better outcomes.

As training continues, the agent gains more experience and updates its policy. After approximately **500 episodes**, the rewards become stable. This indicates that the agent has learned a consistent strategy.

Stable rewards suggest:

1. The agent has reduced unnecessary exploration.
2. The policy has become more consistent.

3. The agent is repeatedly selecting effective actions.
4. Learning is approaching convergence.
5. Performance is becoming more robust to environmental randomness.

In a stochastic environment, some variation may still remain even after convergence because the environment itself contains randomness.

Conclusion: Stabilization after **500 episodes** indicates that the agent has improved its policy, learned a reliable strategy, and is approaching convergence. The stable reward pattern demonstrates **robust and consistent learning performance**.

Code of Conduct:

*I G Saswanth (192425150)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: G Saswanth

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure,	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

		visuals, and terminology			
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